

Markscheme

Mathematics: analysis and approaches
Practice question bank

120. (a)

5-digit numbers greater than 30 000 are formed using the digits 1, 2, 3, 4, 5, 6, with no digit repeated.

Find the number of such 5-digit numbers.

the first digit must be 3, 4, 5 or 6 for the number to exceed 30 000: 4 choices

(M1) **A1**

the remaining 4 digits are arranged from the 5 remaining digits: ${}^5P_4 = 120$

(M1) **A1**

multiply: 4×120

(M1)

$= 480$

A1

[6 marks]